

TABLE II
Some Aspects of Designing for Assembly and its Impact

DFA aspects	Impact
PCB design for assembly	Some of the important parameters that impact the assembly are: • Use of SMT and through-hole technologies • Mechanical fasteners and their positioning in the PCB • Shape and size of the PCB assembly • Cable assembly design and connector design
Electronic packaging design, including cable and connector selection	• A common mistake is committed while selecting cables and connectors, as this is done without understanding the specific use of the cable and connector in the design • Provisioning of the fastener in the cabinet to help easy assembly without any special jig or fixtures

TABLE III
Manufacturing Aspects and Their Impact on the Product

Manufacturing aspects	Impact
Number of types of parts	While there is no restriction on the number of types of parts, one key issue is that, as the number of unique parts increase, the machines that are used for assembling the components have to be programmed several times, increasing the cost and time
Usage of correct parts	With the size of SMT parts shrinking by the day, most designers tend to use the smallest part in their design, irrespective of the fact whether the design requires it or not, leading to unexpected problems

The test strategy for a product is expensive and this depends on various factors, the most important being the volume of the product. With higher volume, the test methodology has to be fast and reliable. Typically, contract manufacturers use in-circuit testers (also known as bed of nail testers). In this test, for each PCB, a dedicated jig with pogo pin nails is used, which is connected to the test equipment that drives special test vectors to test each device in the PCB. These testers are capital intensive and are suitable for large-volume products as the testing cost can be easily recovered. However, to get the best results, designers have to provide test points in their design to ensure the testers test the PCB entirely. Unless development involves both the test engineering team and the design team, good test coverage of the boards can be an issue.

When the product volumes are low, the biggest challenge is how the PCBs and the product will be tested. This is an area where designers have to be very careful, as in low volume production, often contract manufacturers depend on the design team to provide

them with dedicated testers or test software, which will test PCBs without the need of expensive test jigs.

Most designers are ignorant of the fact that a simple addition to their design can make the boards testable up to 100 per cent, thus avoiding the expensive test jig. With the availability of test engineering tools like LabView, Agilent VEE, etc, the cost of testing can be brought down to minimum. With the judicious combination of hardware and firmware, most of the PCBs can be made testable 100 per cent. So when the designers start a new design, one of the key aspects is the ability to test the board quickly and with least expense.

Design for support (DFS)

Finally, one of the least addressed aspects of low-cost design is the ability to support the product when it is in use. This includes everything—from rectifying the fault when reported, to upgrading the software if needed, for better features.

This becomes crucial when the product life is long, especially in domains like telecom, avionics and

Designing products, specifically keeping in mind the process of manufacturing (DFx), helps in controlling the cost

medical products that are normally in use for 15-20 years. Designers need to be cognizant of the fact that these products need long-term support and the design has to incorporate support aspects both in the hardware and software.

In the hardware, the support issue can come from how the system hardware is partitioned, systems engineering aspects like cabling, connectors, etc. While in software, this starts from offline diagnostics to boot-loading features for quick software upgrade.

Of late, one of the features that most devices have is diagnostics over the internet or remote diagnostics, which sometimes requires hardware features like GPS sensors incorporated into the product so that the service personnel can reach the location quickly. Unless these support features are planned well in advance during the design phase itself, adding them once deployed is very difficult, and a product without good support features can involve very high after sales support costs, in turn, leading to product failure.

So we have found how DFx can effectively reduce the time to reach the market and at the same time save product costs, leading to savings to the product sellers. We have also seen some of the important aspects of DFx and how designers have to be cognizant to the fact that DFx is an integral part of their design cycle, especially in the outsourced manufacturing scenario. While the issues discussed may not be exhaustive, they do highlight the importance of adopting DFx principles early in the design cycle. ●

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